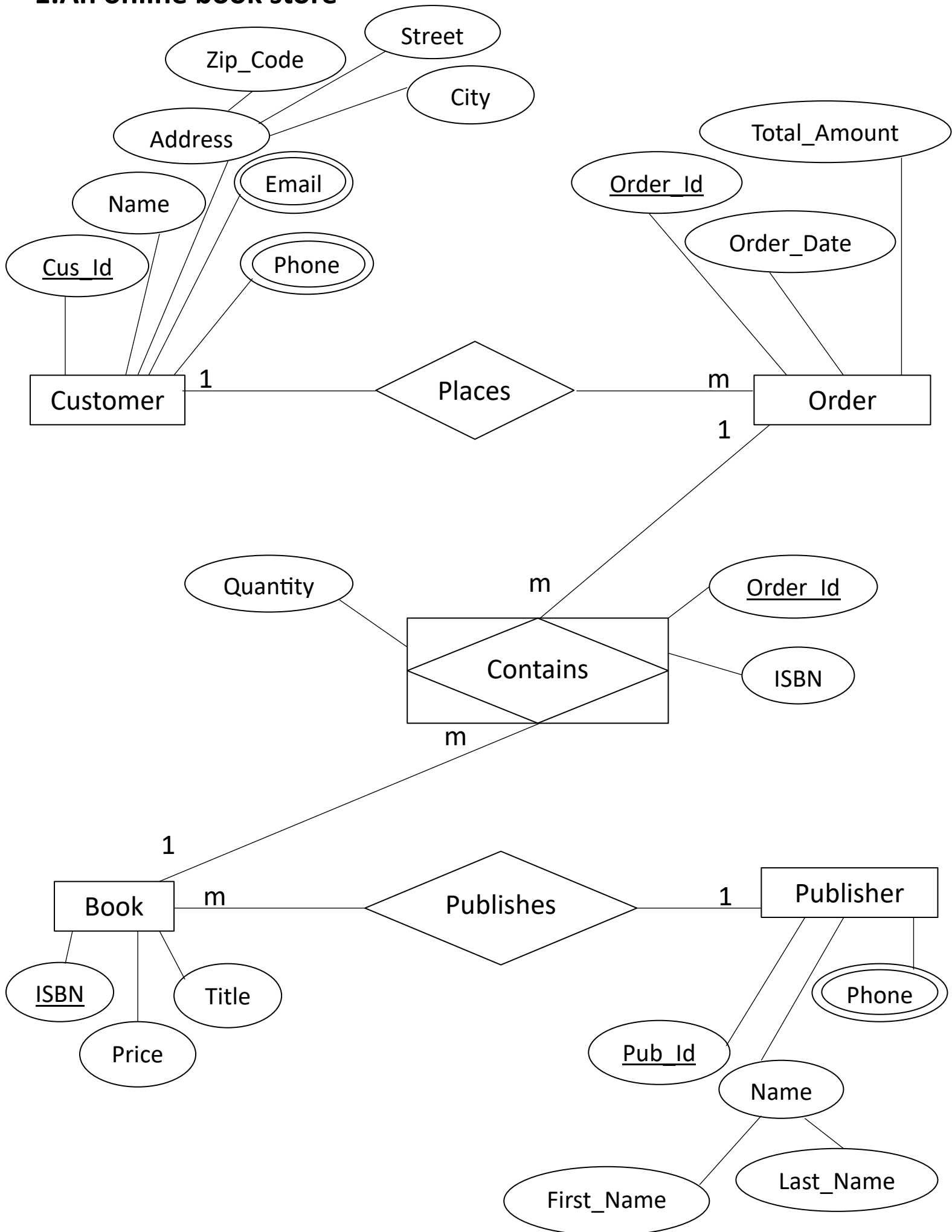


Assignment 02 - ER Diagram

1. An online book store



a)Types of entities

- **Customer** : is a strong entity. It's primary key is Cus_Id.
- **Book** : is a strong entity. It's primary key is ISBN.
- **Order** : is a strong entity. It's primary key is Order_Id.
- **Publisher** : is also a strong entity. It's primary key is Pub_Id.
- **Contains** : is a composite entity, must contain the primary keys of the associated entities as it's foreign keys.

b)Types of attributes

❖Customer

- **Cus_Id, Name** →Single value attributes
- **Address** →Composite attribute
 - Address can be subdivided into smaller sub parts like Street, City, Zip_Code.
- **Phone, Email** →Multi-valued attributes
 - One customer can have one or more phone numbers and emails.

❖Order

- **Order_Id, Total_Amount, Order_Date** →Single value attributes.

❖Book

- **ISBN, Title, Price** →Single value attributes

❖Publisher

- **Pub_Id** →Single value attribute
- **Phone** →Multi-valued attribute
 - One publisher can have more than one phone numbers.
- **Name** →Composite attribute
 - Name can be subdivided into sub parts like First_Name, Last_Name.

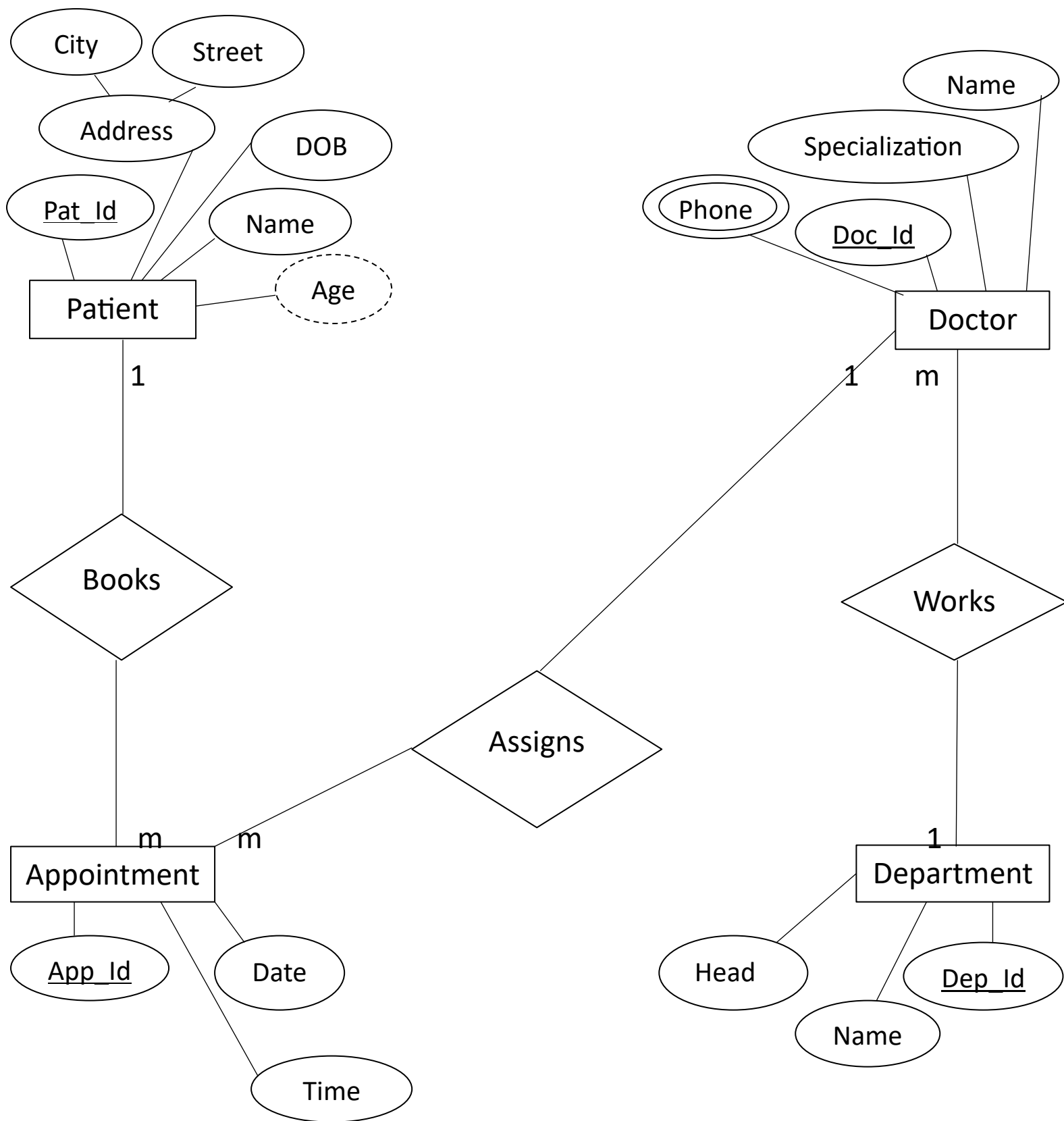
c) Cardinalities

Relationship	Cardinality	Reason
Places	One to many	One customer can place many orders but one order definitely belongs to one customer.
Publishes	One to many	One publisher can publish multiple books and one book belongs to one publisher.
Contains	Many to many	One order can have multiple books and one book can have multiple orders. That relationship creates composite entity which is contain primary keys of Book and Order. It has also attribute like Quantity.

d) Relational Schema

- Customer(Cus_Id, Name, Zip_Code, Street, City)
- Customer_Phone(Cus_Id, Phone)
- Customer_Email(Cus_Id, Email)
- Order(Order_Id, Cus_Id, Total_Amount, Order_Date)
- Book(ISBN, Pub_Id, Title, Price)
- Publisher(Pub_Id, First_Name, Last_Name)
- Publisher_Phone(Pub_Id, Phone)

2.An Online Hospital Management System



a)Types of entities

- **Patient** : is strong entity. It's primary key is Pat_Id.
- **Doctor** : is a strong entity. It's primary key is Doc_Id.
- **Appointment** : is a strong entity. It's primary key is App_Id.
- **Department** : is a strong entity. It's primary key is Dep_Id.

b)Types of attributes

❖Patient

- **Pat_Id, Name** →Single value attribute.
- **Address** →Composite attribute.
 - Address can be subdivided into smaller sub parts like Street, City.
- **DOB** →Stored attribute.
- **Age** →Derived attribute.
 - Age can be calculated by DOB attribute.

❖Doctor

- **Doc_Id, Name, Specialization** →Single value attribute.
- **Phone**→Multi-valued attribute
 - One doctor can have one or more phone numbers.

❖ Appointment

- **App_Id, Date, Time** →Single value attributes.

❖ Department

- **Head, Name, Dep_Id** →Single value attributes.

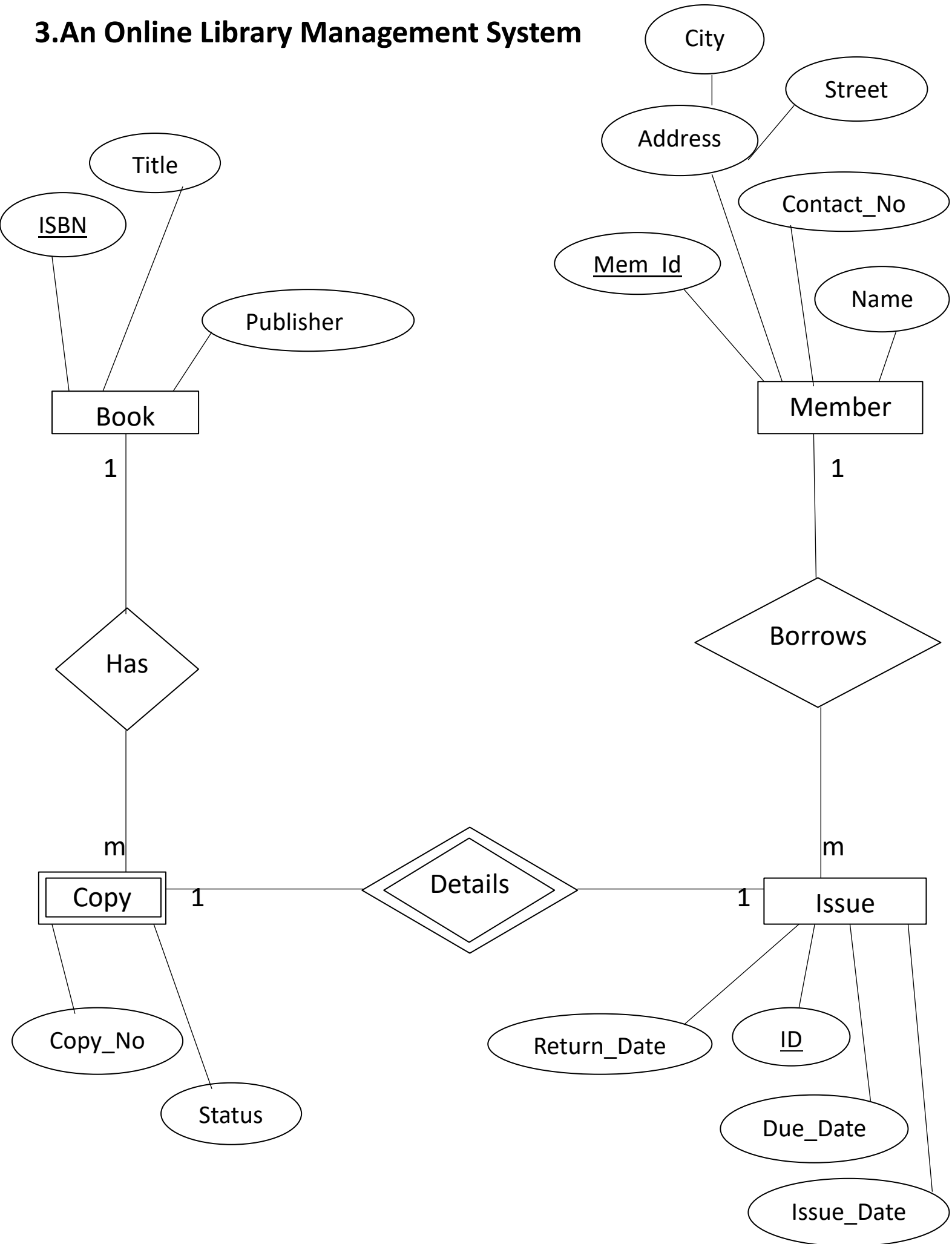
c) Cardinalities

Relationship	Cardinality	Reason
Books	One to many	One patient can have many appointments but one appointment definitely belongs to one patient.
Assigns	One to many	While one doctor can hold multiple appointments, each appointment is held by only one doctor.
Works	One to many	One department can have many doctors working in it, but each doctor works in one department.

d) Relational Schema

- Patient(Pat_Id, Street, City, Name, DOB, Age)
- Doctor(Doc_id, Specialization, Name)
- Doctor_phone(Doc_Id, Phone)
- Appointment(App_Id, Pat_Id, Doc_Id, Date, Time)
- Department(Dep_Id, Head, Name)

3.An Online Library Management System



a)Types of entities

- **Member** : is a strong entity. It's primary key is Mem_Id.
- **Book** : is a strong entity. It's primary key is ISBN.
- **Copy** : is a weak entity. It has no primary key.
- **Issue** : is a strong entity. It's primary key is Issue_Id.

b)Types of attributes

❖Member

- **Address** → Composite attribute
 - Address can be subdivided into smaller sub parts like Street, City.
- **Contact_No, Name, Mem_Id** → Single value attributes.

❖Book

- **ISBN, Title, Publisher** → Single value attributes.

❖Issue

- All are the single value attributes.

c)Cardinalities

Relationship	Cardinality	Reason
Has	One to many	A book can have multiple copies and each copy belongs to a specific book.
Borrows	One to many	One member can have multiple issues, but one issue belongs to only one member.
Details	One to one	One issue is the release of one copy and one copy is associated with a specific issue.

d)Relational Schema

- Member(Mem_Id, Name, City, Street, Contact_No)
- Book(ISBN, Title, Publisher)
- Issue(ID, Mem_Id, Copy_No, ISBN, Due_Date, Issue_Date, Return_Date)